

Replace the four point bearing and adjust the gaps at the infeed drive and infeed screw

SI 163.2003en

Version 1.2

1 Disassembling the infeed drive

Unlock the two locking levers on the hopper housing and carefully tip the hopper backwards.

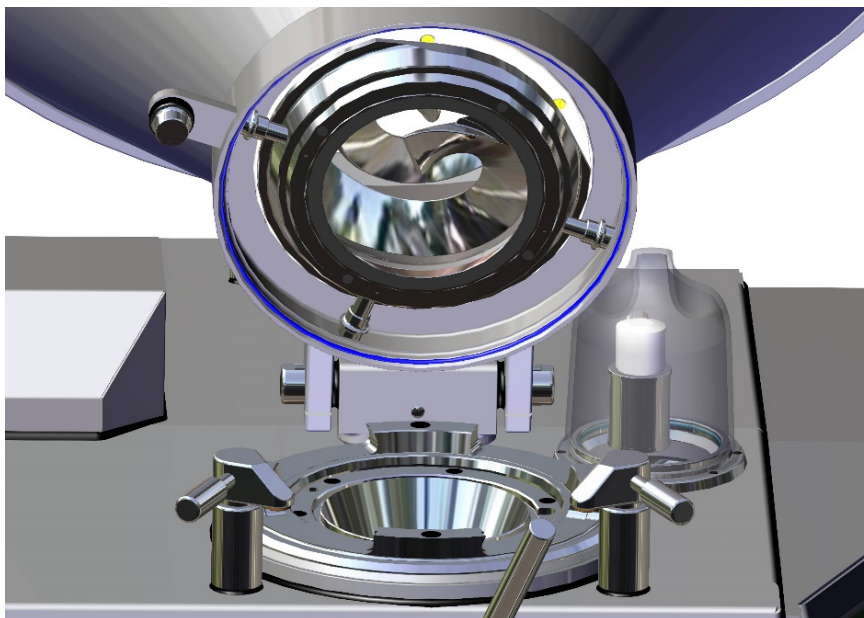


figure 1

Remove the scraper in the hopper (if present). Push back the three sliding sleeves in the hopper flange, holding the infeed screw steady with the other hand.



figure 2

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Carefully twist the infeed screw out of the hopper and take out the blue sealing rings.

i *Make sure the black colored sealing surface (upper and lower side at the flange) will not be damaged while doing this.*

ATTENTION!

In case the surfaces are damaged, the blue sealing rings will not last for a long lifetime and the complete infeed screw needs to be replaced, as it's not possible to repair the sealing surface.



figure 3

Remove the plastic caps from the catch ring and the four cylinder bolts.

ATTENTION!

While assembling the catch ring, later the cylinder bolts will need to tighten evenly crosswise with a torque of $30 + 5 \text{ Nm}$.

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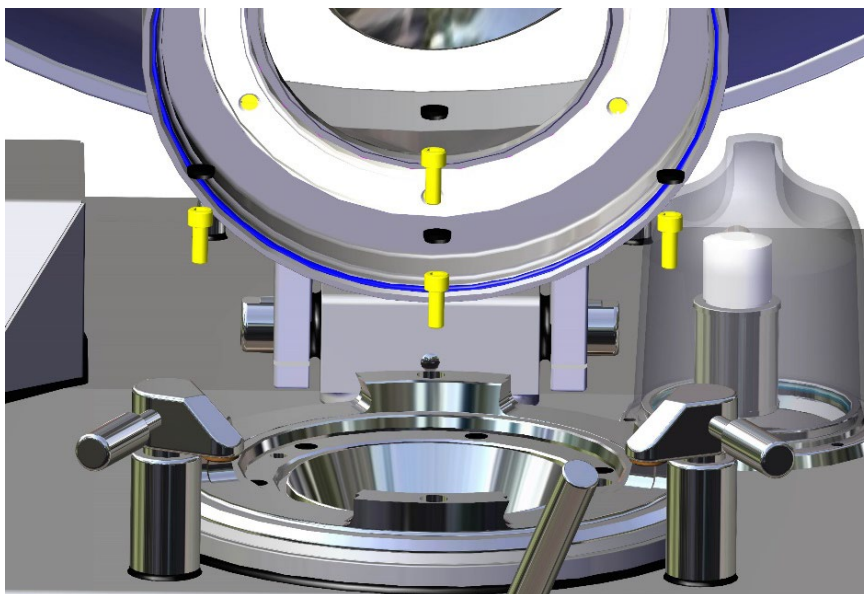


figure 4

Carefully remove the catch ring.

i Four cylinder bolts (e. g. M12 x 40 mm) can be used as jack screws, if needed.

ATTENTION!

Make sure the black colored sealing surfaces (lower side at the catch ring) will not be damaged while doing this. In case the surfaces are damaged, the sealing rings will not last for a long lifetime. And the complete catch ring needs to be replaced, as it's not possible to repair the sealing surface.

Additionally the catch ring should have only a parallel run out of 0.05 mm.

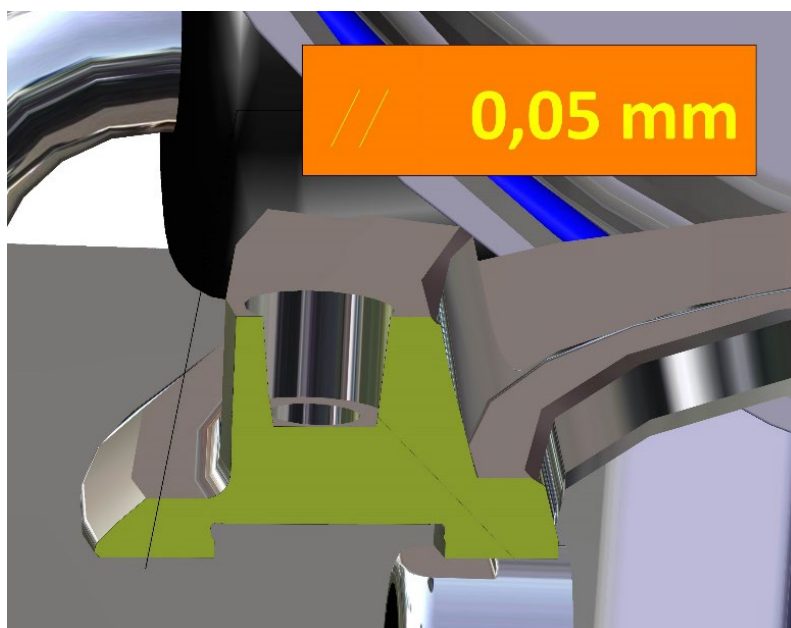


figure 5

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Remove the sealing rings.

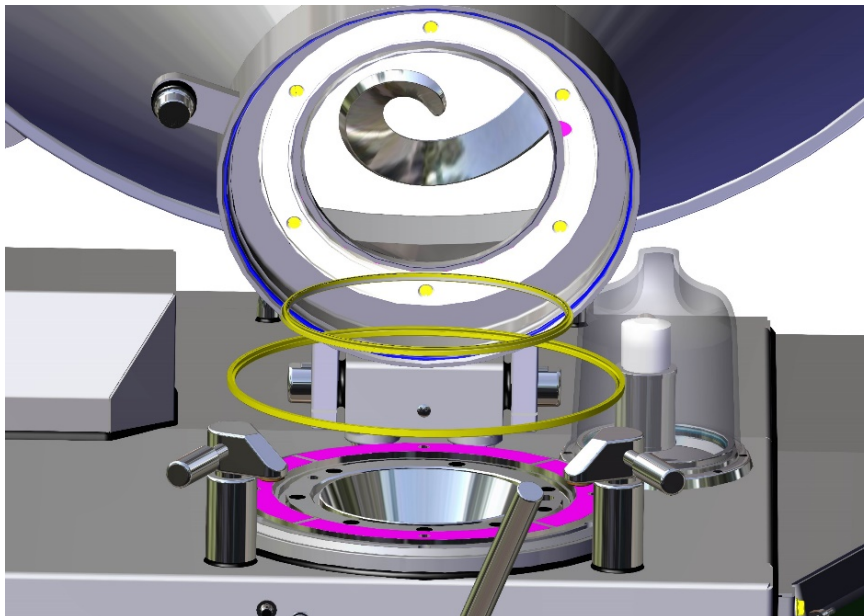


figure 6

Remove the shims at the outer ring from the four point bearing (if present).

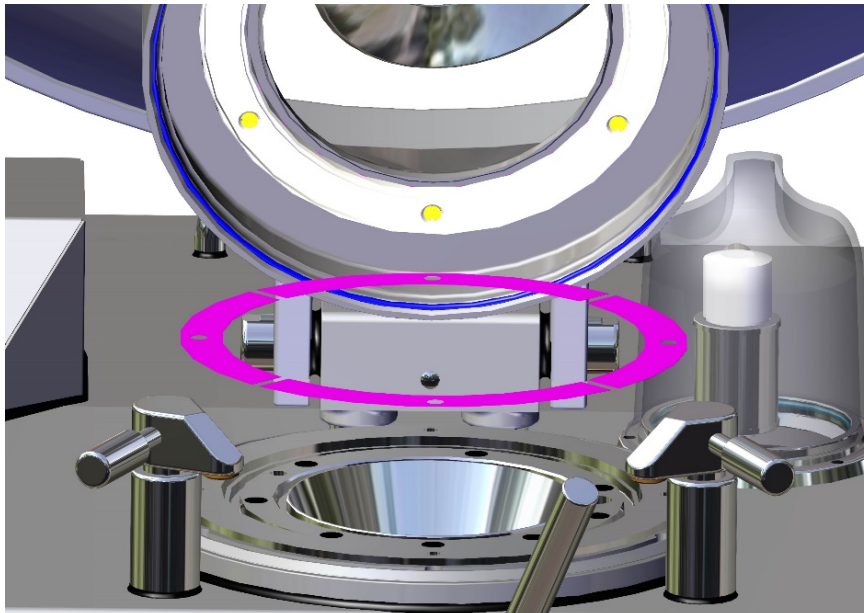


figure 7

Remove the sealing rings underneath the catch ring.

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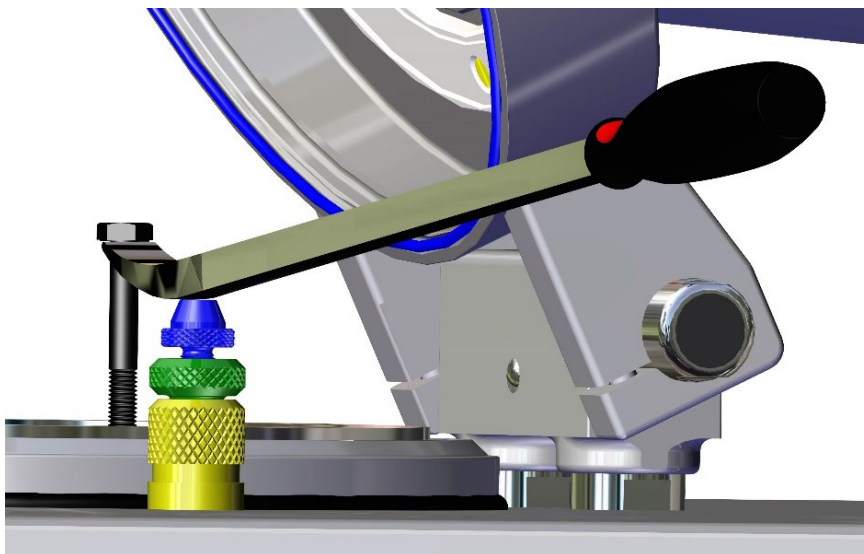


figure 8

To check the wear at the four point bearing, the outer ring need to be lifted carefully.

Therefore screw in one hexagon bolt (e. g. M10 x 70 mm) into one of the four inner threads from the outer ring of the bearing. Place a threaded screw jack (e. g. a piece of metal, or plastic) as a counter bearing close to the hexagon bold. Careful position a pry bar underneath the head of the bolt and push downward the handle of it.

While doing this, the grease around the four point bearing is not allowed to move.

If so, the bearing is worn, or damaged and need to be replaced.

Remove the plastic caps and the eight cylinder bolts.

i While assembling later the hopper inlet, these screws must be tighten evenly cross-wise.

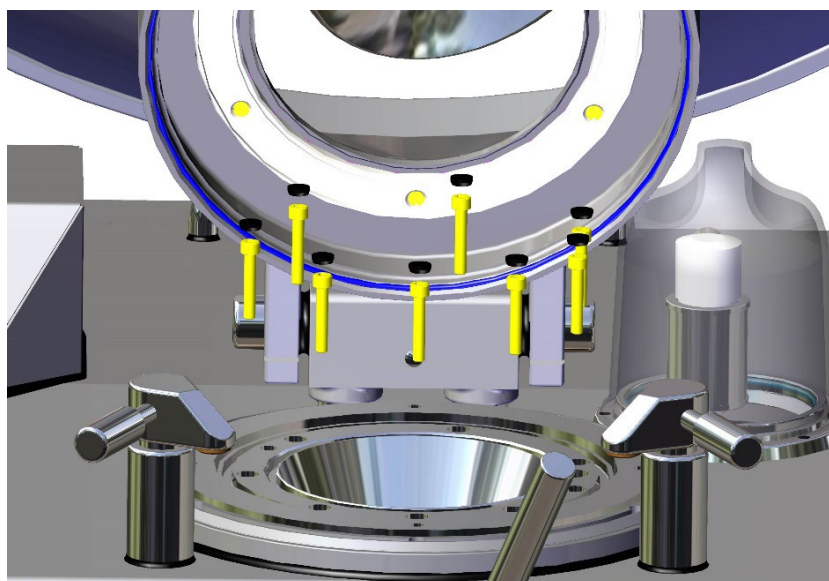


figure 9

Replace the four point bearing and adjust the gaps at the infeed drive and infeed screw

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Screw in two hexagon bolts (e. g. M10 x 100 mm) in black steel and use them as jack screws to disassemble the hopper inlet.

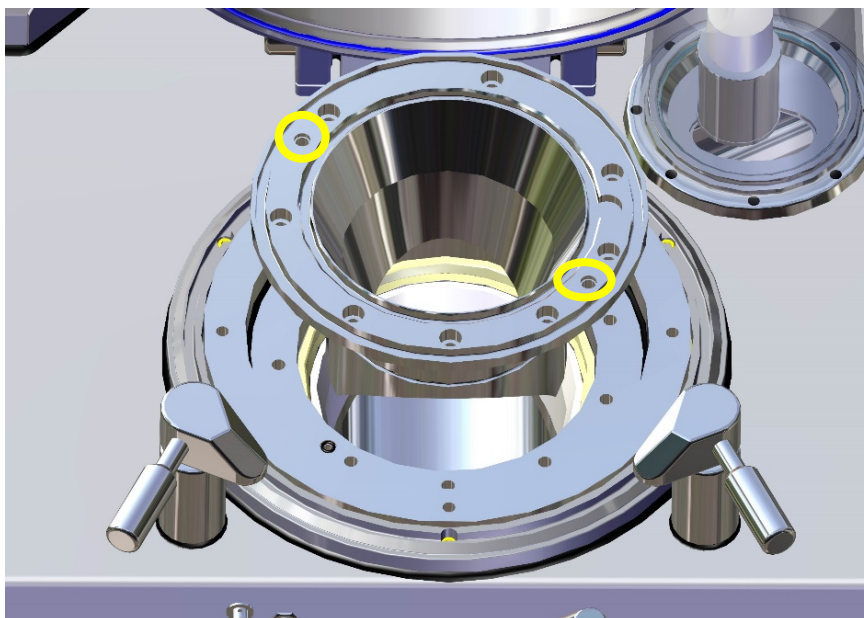


figure 10

Remove the four point bearing and watch out for the three o rings. They will need to be replaced when assembling the ring gear again.

Visual inspect the sprocket from the unit and watch out for wear and damages.

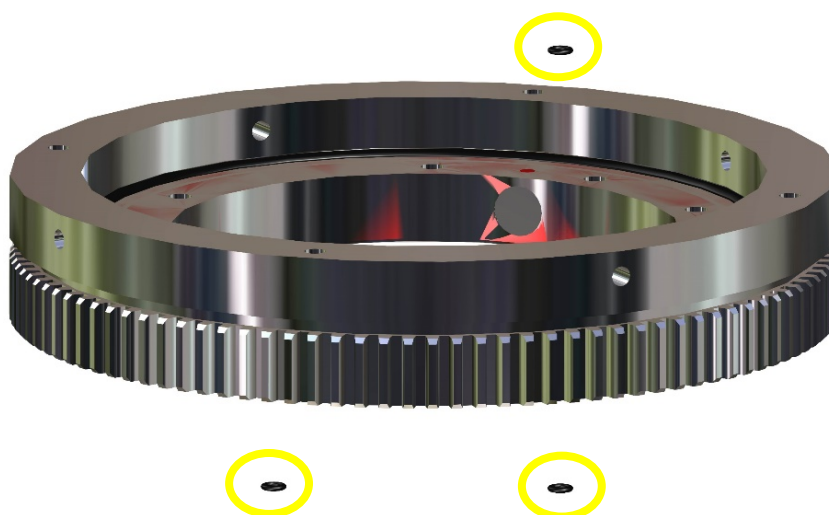


figure 11

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Loosen the six flat head cylinder bolts at the lower outer ring.

These bolts are special, as they have a center drilling into the flat head to guide a special Allen key. Make sure you'll have an appropriate tool with you.

In case a standard Allen key will also make it, but it's not guided and easily the flat head can be damaged.

i While assembling later the lower outer ring, these screws must be hand tighten evenly crosswise.

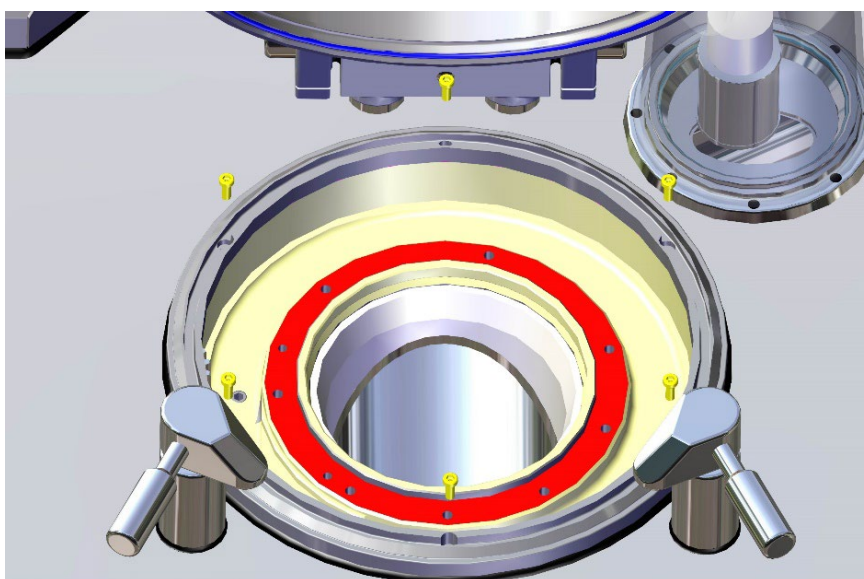


figure 12

Undo the lower, outer ring, the packing cord and the shims.

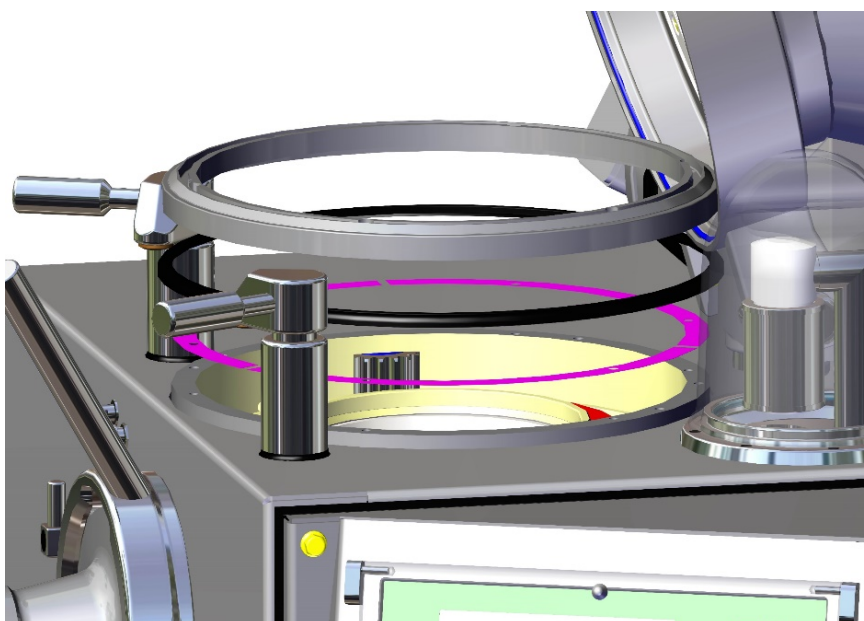


figure 13

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Careful clean the grease area (yellow) and check the surface (red) from the inner ring of the four point bearing.

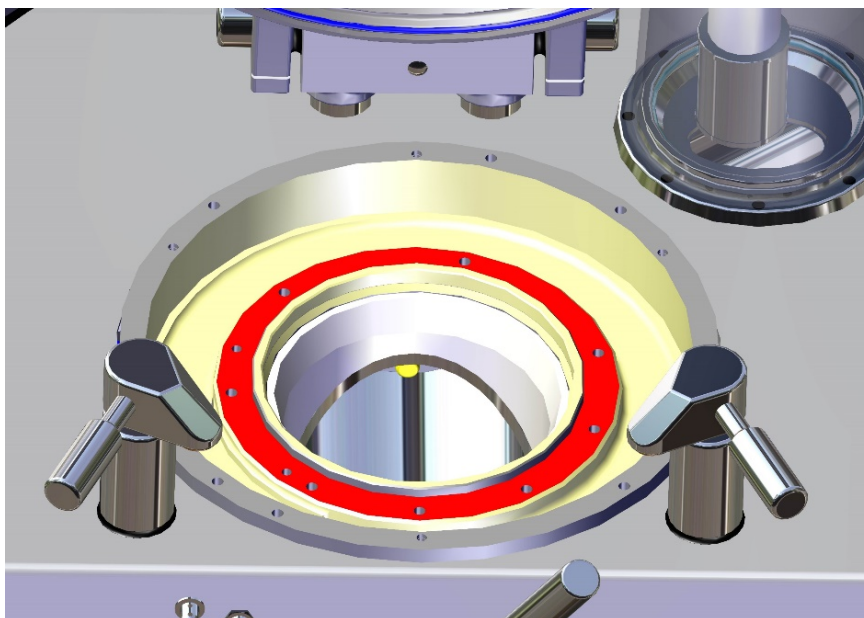


figure 14

Visual inspect the pinion from the infeed drive and watch out for wear and damages.

In case of wear or damages, the pair (pinion and ring gear) need to be replaced.

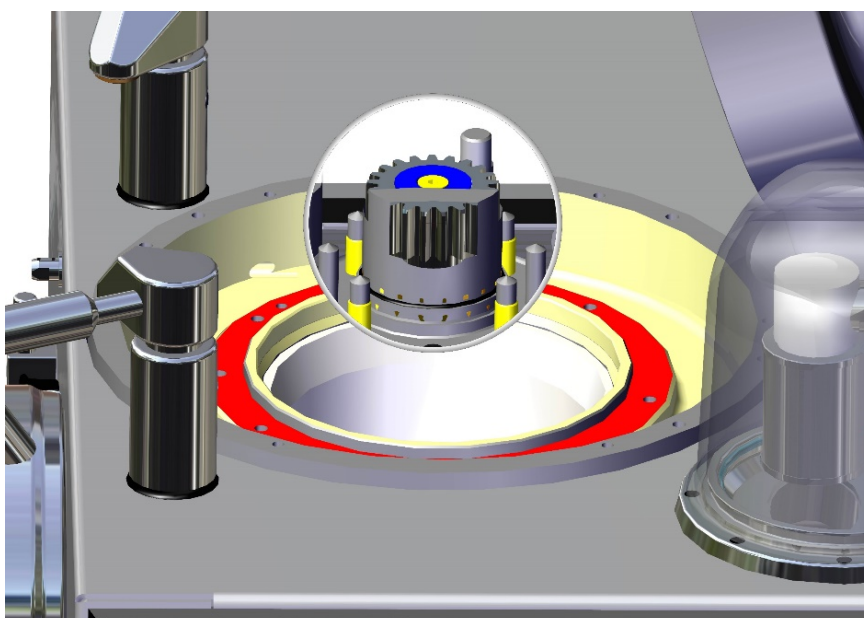


figure 15

2 Assembling the infeed drive

Start assembling of the unit from here in reverse direction. Clean the filling cylinder with cold solvent and degrease the assembly area with a brake cleaner before assembling the hopper inlet.

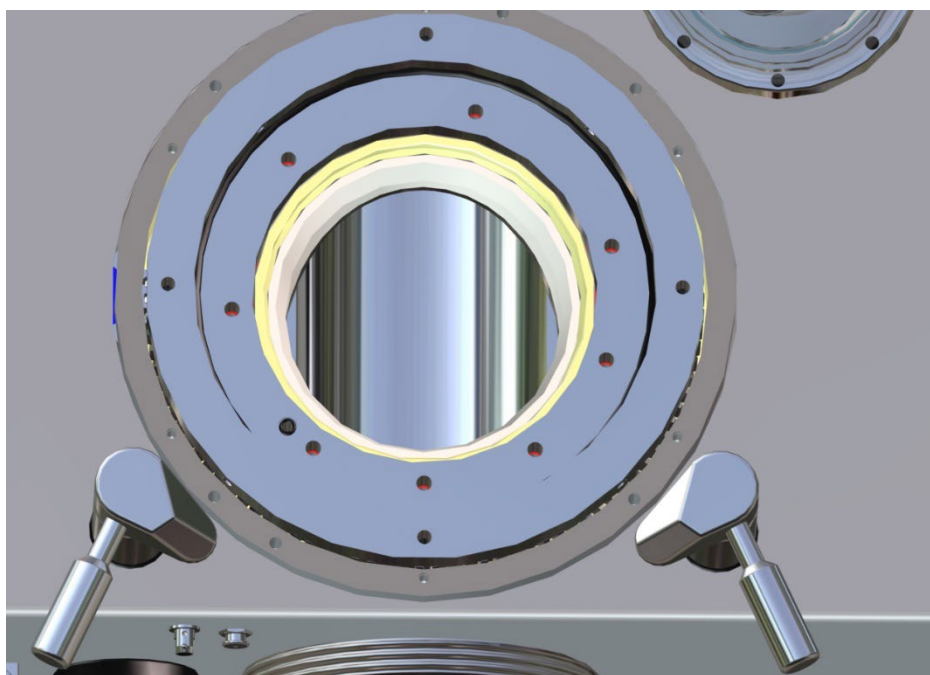


figure 16

Apply below the assembled O-ring at the hopper inlet a circumferential seam with a width of 8 to 10 mm with a fast-curing, elastic two-component adhesive and sealant (e. g. Weicon Flex 310 M 2K).

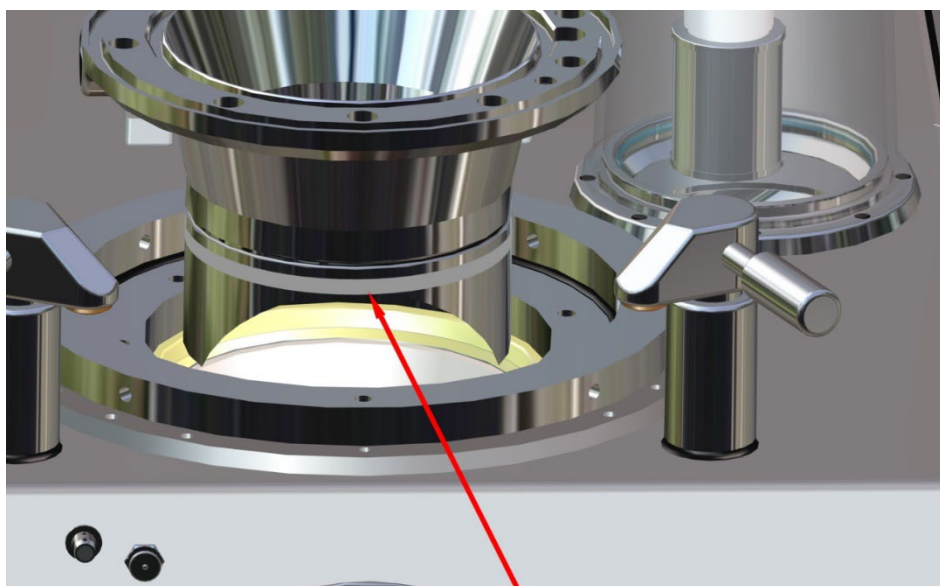


figure 17

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Prior to the assembling it should be ensured that no adhesive will get to the front surface of the hopper inlet.

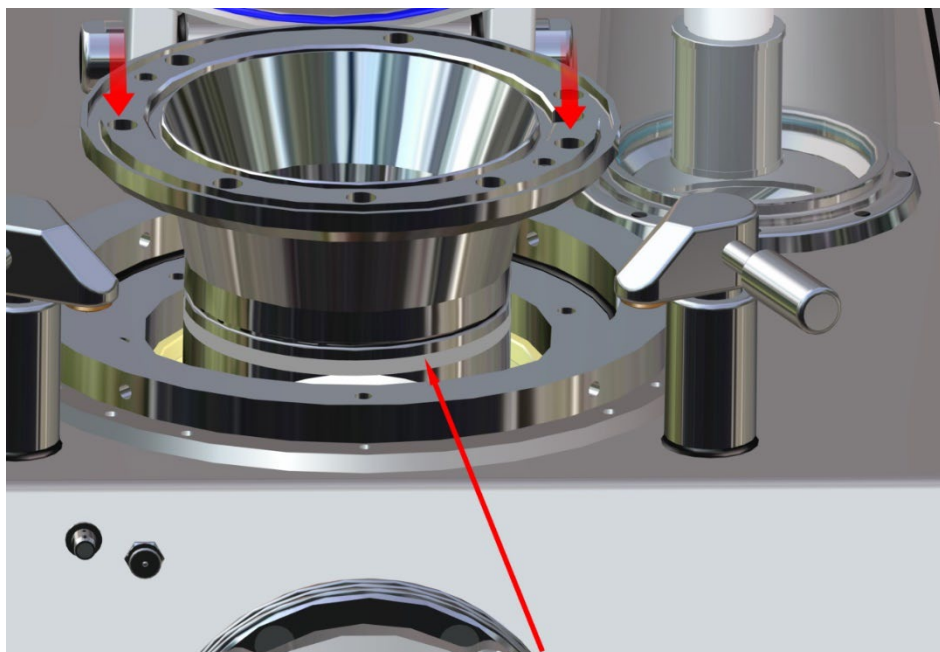


figure 18

After the assembling process has been successful completed all transitions to the filling cylinder will need to be checked. Use a feeler gauge to control if there aren't any adhesive residues at the cleavage site.

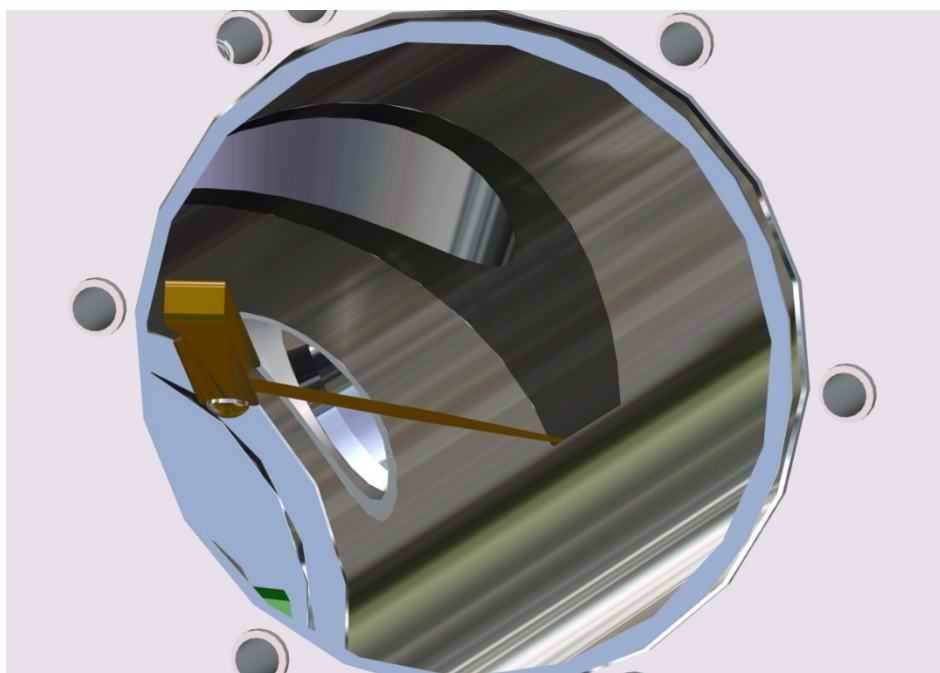


figure 19

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Assemble the catch ring, but without the sealing rings and the shims to check the air gaps underneath the catch ring.

It's recommended to fill the area between the filling cylinder and the four point bearing with the VEMAG HIGH PERFORMANCE grease, before assembling the four point bearing. Approx. one to one and half cartridges will be needed.

ATTENTION!

make sure the area won't be overfilled and the inner ring of the four point bearing will sit directly on the filling cylinder.

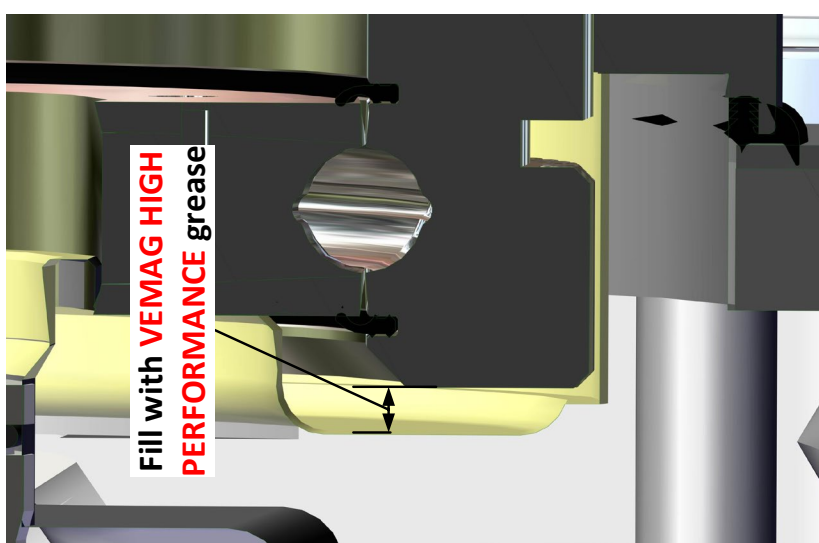


figure 20 - Sectional view: four point bearing/filling cylinder

Assemble the catch ring without the appropriate sealing.

ATTENTION!

the four cylinder bolts will need to tighten *evenly crosswise* with a torque of 30 + 5 Nm.

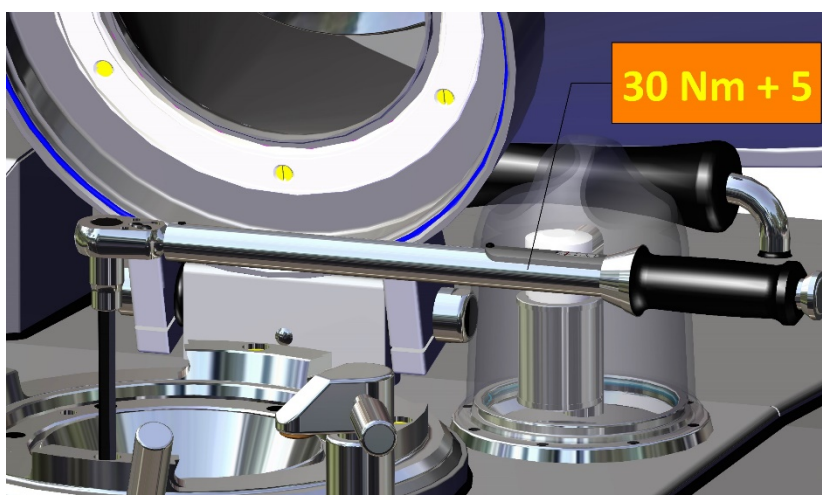


figure 21

Replace the four point bearing and adjust the gaps at the infeed drive and infeed screw

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ATTENTION!

Check the air gap of 0.25 ± 0.05 mm with a feeler gauge, between the catch ring and the hopper inlet. Measure at different positions. In case the air gap isn't in tolerance, it needs to be adjusted.

Therefore two sets of shims will be available:

1. 142300340 (0.05 mm)
2. 142300350 (0.10 mm)

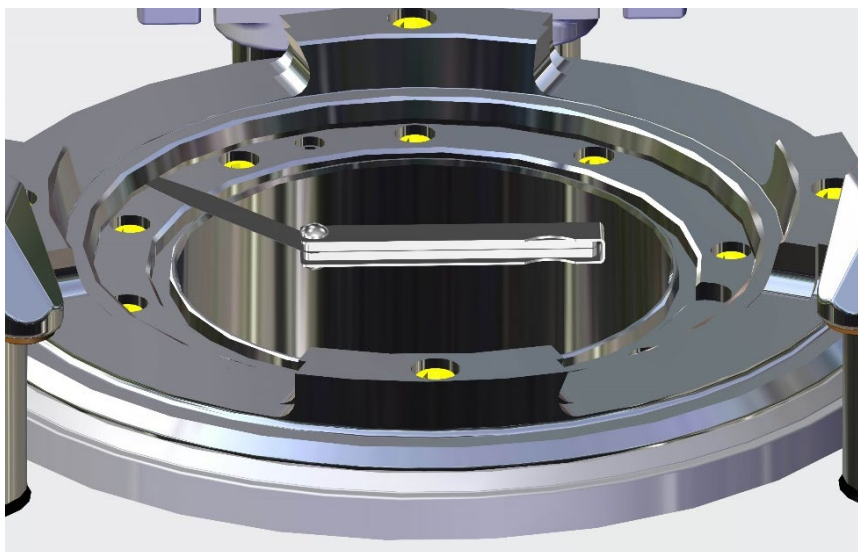


figure 22

ATTENTION!

Check the air gap of 0.25 ± 0.05 mm with a feeler gauge, between the catch ring and the lower, outer ring. Measure at different positions. In case the air gap isn't in tolerance, it needs to be adjusted.

Therefore two sets of shims will be available:

1. 142300470 (0.05 mm)
2. 142300480 (0.10 mm)

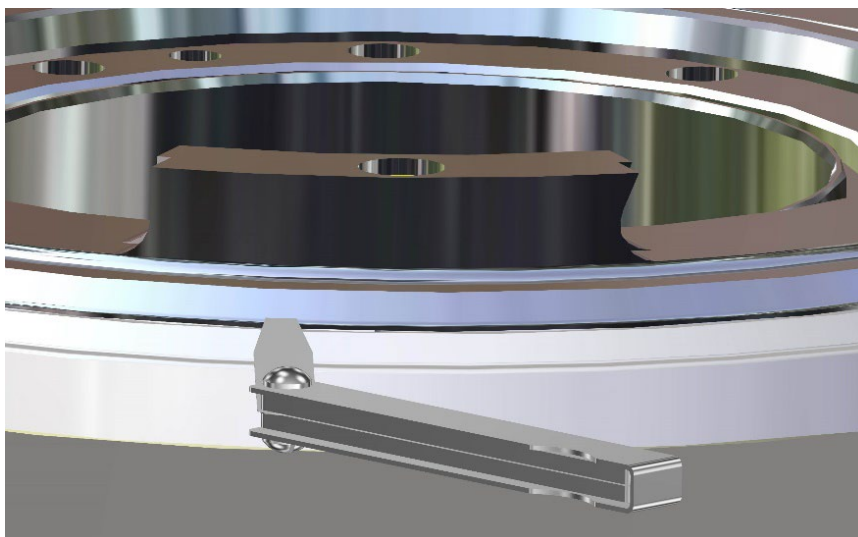


figure 23

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Disassemble once again the catch ring and install the sealing ring at the hopper inlet. Make sure no dirt or grease is inside the groove.

Use a hammer handle (or similar) to push down the sealing ring into the groove. Repeat this at different positions. When removing the hammer, the sealing ring must come off the groove a little.

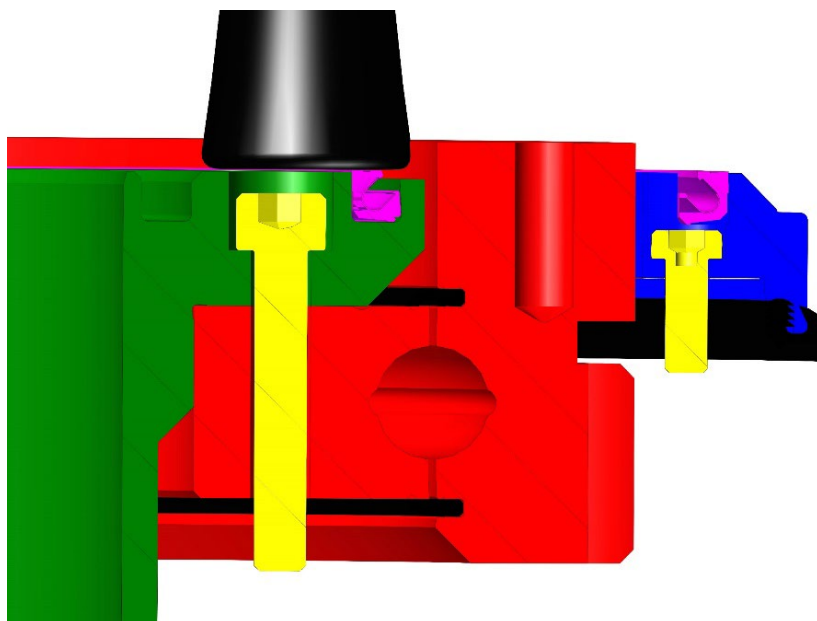


figure 24

Repeat the same procedure with the sealing ring at the lower, outer ring.

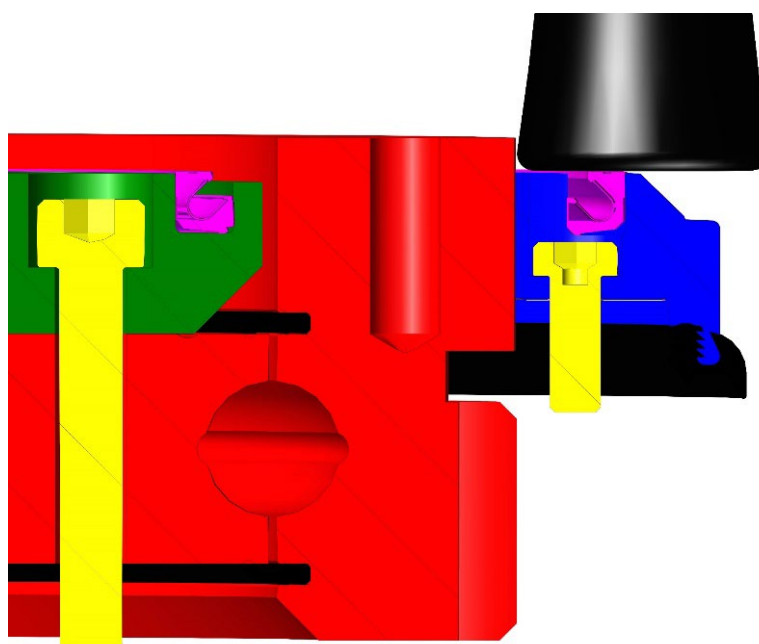


figure 25

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ATTENTION!

both sealing rings must be fitted in such a way that the spring is visible at the top.

In case one or both sealing rings stuck into the groove, analyse the problem.

When the outer diameter will not match, careful scrape off material by using a cutter knife.

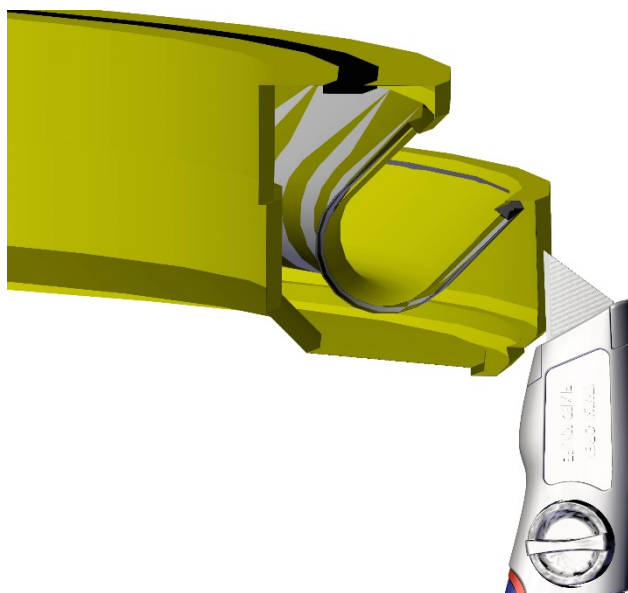


figure 26 - Outer diameter of the sealing ring

When the inner diameter will not match, careful scrape off material by using a cutter knife.

Alternative other, new sealing rings may fit into the grooves without doing this procedure.

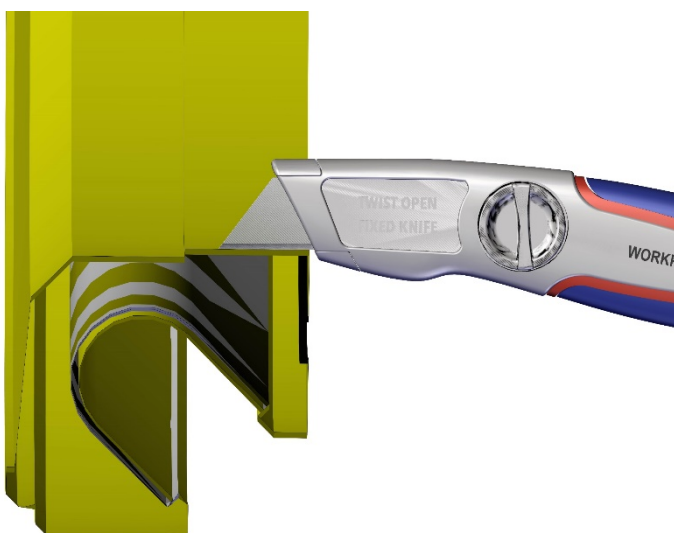


figure 27 - Inner diameter of the sealing ring

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When all air gaps are in tolerance assemble the infeed.

Install once again the infeed screw, but without the blue colored sealing ring and tip forwards the hopper and lock the two locking levers on the hopper.



figure 28

ATTENTION!

Lean over the closed hopper and check the air gap of $0.1 + 0.05$ mm with a feeler gauge, between the sliding ring and the upper flange of the infeed screw.



figure 29

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In case the air gap isn't in tolerance it needs to be adjusted. Therefore loosen the six countersunk screws and remove the shims underneath.

There are two sets of shims available:

1. 142300410 (0.05 mm)

2. 142300420 (0.10 mm)

ATTENTION!

The countersunk screws will need to tighten evenly crosswise with a torque of 5 Nm.

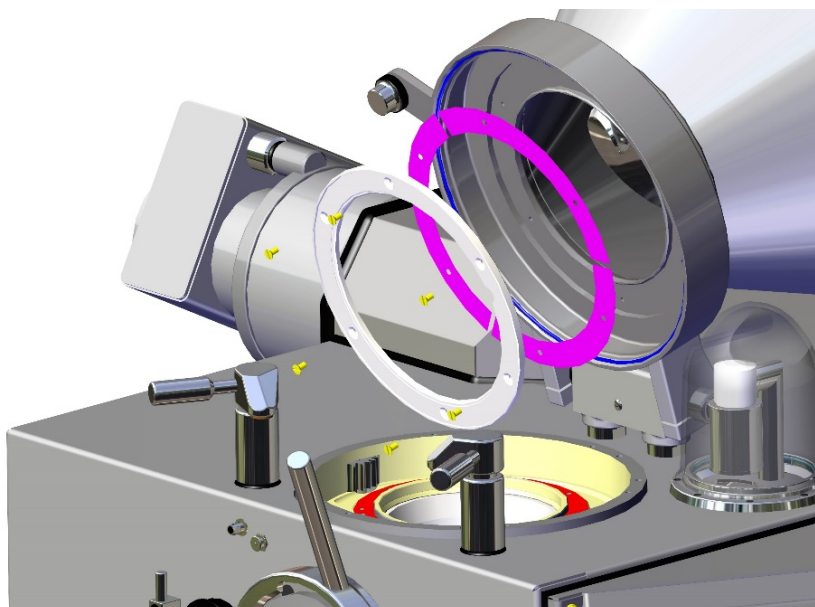


figure 30

Assemble all plastic covers and insert all sealing rings like the machine would be prepared for the production.

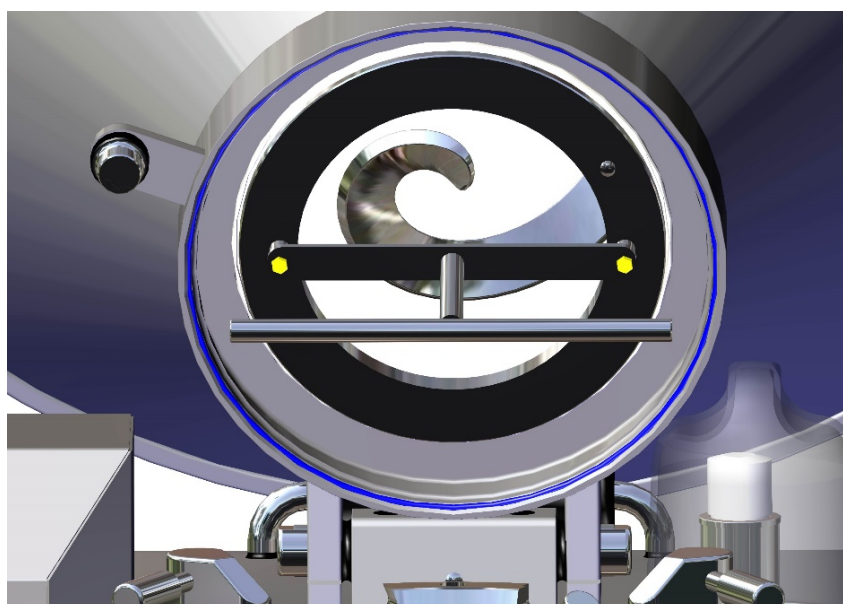


figure 31 - Dairy version

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At the Dairy version you'll need to use the mounting tool (148031000; comes with the machine) to assemble/remove the sliding ring. Set the mounting tool with the pins into the bores of the sliding ring and turn it counter clockwise in order to unlock the ring. Carry out the mounting in reverse order, but make sure the o ring is in position.

The clearance are equal to the standard design of the sliding rings. (Refer to figure 29)

On the end the infeed drive system need to be filled with the VEMAG HIGH PERFORMANCE grease.

Therefore chose a program in straight filling mode, select knee lever II and a speed of 20 %. Push the knee lever once and lubricate with the grease gun (provided by VEMAG) and press in until new grease comes out of the relieve valve.

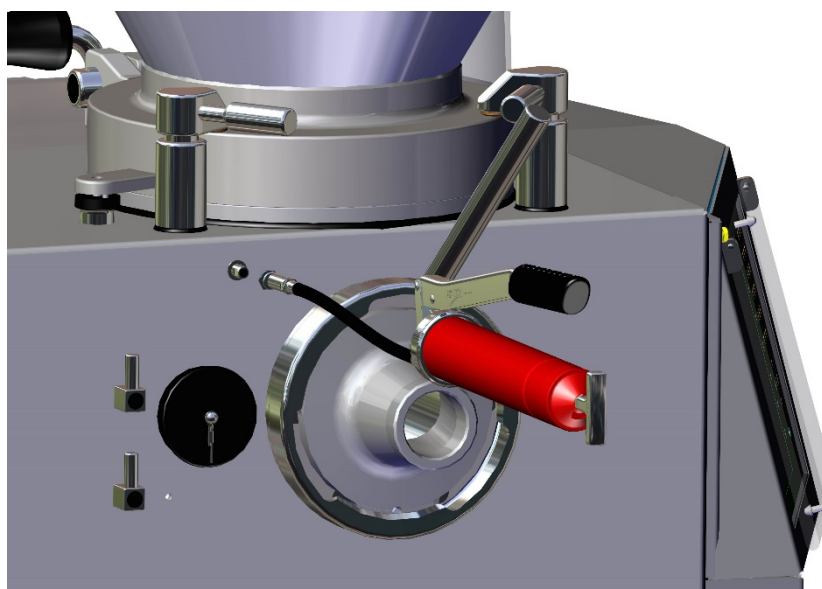


figure 32